

RABAB KHAN RONGON

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RESEARCH INTEREST

Human Computer Interaction, Artificial Intelligence, Machine Learning, Software Test Engineering.

ACADEMIC CREDENTIALS

M.Sc. Military Institute of Science and Technology, Dhaka, Bangladesh *April 2024 - Continuing*
Department: Computer Science and Engineering || Major: Artificial Intelligence || CGPA: 3.50
Selected Course Work: Advanced AI, Advanced ML, Neural Networks, Data Mining, Fuzzy Logic, HCI

B.Sc. International University of Business Agriculture and Technology, Dhaka, Bangladesh
Department: Computer Science and Engineering || CGPA: 3.21 out of 4.00 *Graduation: Jan 2024*
Selected Course Work: OOP in C++, Data Structure & Algorithm, DBMS, Software Engineering, System Design, Theory of Computation, Computer Architecture, Operating Systems, Microprocessor.

PUBLICATIONS

Undergraduate Thesis

[1] ***Implementation of Energy Efficiency Cloud Data Center Architecture Model using Green Cloud Simulator for National Data Center of Bangladesh*** ☐
Supervised by: Krishna Das ☐

Journal Paper

[2] **Rabab Khan Rongon**, Krishna Das, “***An Energy Efficient Cloud Data Center Architectural Model for the National Data Center of Bangladesh***” in "Journal of Discrete Mathematical Sciences and Cryptography", Volume 27, Issue 6, Page 1905-1917 (Scopus Indexed Q1 Quartile, Cite Score: 3.1) ☐

[3] M.A Eti, **Rabab Khan Rongon**, “***Investigating tRNA-Derived Small RNA Expression Patterns and Bioinformatics Analysis in Epicardial Fat of Atrial Fibrillation Patients***” in International Journal for Research in Applied Science and Engineering Technology ☐

Book Chapter

[4] A. Islam, S. Zaman, **Rabab Khan Rongon**, S.T. Sharif, H. Jahan, “***An Attention-Driven Deep Learning Model for Predicting Human Life Expectancy***” in 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025). (Accepted)

Conference Paper

[5] **Rabab Khan Rongon**, M. Akhteruzzaman, S.T. Sharif, K. Das, “***Analyzing Risk Level from RoI and VoI for Robust Data Privacy using Fuzzy Inference System***” in “IEEE 4th International Conference on Robotics, Electrical and Signal Processing Techniques 2025 ☐

- [6] **Rabab Khan Rongon**, S.K. Prangon, K.Das, S.T. Sharif, A. Ashraf, “*Evaluating the Accessibility of Online Pharmacies' Mobile Applications in Bangladesh using Heuristic and Semiotic Analysis*” in 2025 IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking ☐
- [7] **Rabab Khan Rongon**, S.T. Sharif, A. Ashraf, M.N Islam “*Improving CAPTCHA Systems for Better Usability and Accessibility in Mobile Applications*” in IEEE 2nd International Conference on Computing, Applications, and Systems ☐
- [8] S.K. Prangon, **Rabab Khan Rongon**, K.Das, S.T. Sharif, A. Ashraf, “*A Real-Time Automated Traffic Violations Detection System for Motorcycles Using YOLOR*” in 4th IEEE International Conference on Robotics, Automation, Artificial-intelligence and Internet-of-Things. (Accepted)
- [9] **Rabab Khan Rongon**, MS Zaman, S.T. Sharif, S.K Prangon, K.Das “*Deep Learning-Based Recognition of Bangla Sign Language Gestures with CNN, VGG16, and MobileNet*” in 6th IEEE International Conference on Telecommunications and Photonics (ICTP) 2025. (Accepted)
- [10] A. Islam, M. Akhteruzzaman, S. Sultana, **Rabab Khan Rongon** “*Predicting Renewable Energy Usage in Bangladesh Using Environmental and Climatic Indicators: A District-Level Analysis*” in 2026 IEEE 2nd International Conference on Quantum Photonics, AI & Networking. (Accepted)
- [11] **Rabab Khan Rongon**, Krishna Das, “*Obstacle for implementing sustainable computing in Bangladesh and approach to being sustainable through green cloud computing methodology*” in 4th International Conference on Sustainable Computing and Management (SUSCOM-2022) Aug. 23-24, 2022. ☐☐

MANUSCRIPTS UNDER REVIEW

- [12] *Enhancing Maritime Fuel-use Efficiency and Reduction of Carbon Emission: An Experient with Deep Reinforcement Learning Agent* – under review Engineering Applications of Artificial Intelligence
- [13] *Energy-Aware Data Center Management: A Sustainable Approach to Reducing Carbon Footprint* – under review in Discrete Mathematics & Theoretical Computer Science
- [14] *Entropy-Based Profiling of Transcriptomic Organization in Antibiotic-Resistant and Susceptible Escherichia coli* – under review in International Journal of BioAutomation

POSTER PRESENTATION

- [1] M.A Eti, **Rabab Khan Rongon**, “*AI-Driven Early-Warning System for Building Community Health Resilience*” in 5th IEEE International Symposium on Social Implications of Sustainable Technology (SympSIST 2025) ☐
- [2] **Rabab Khan Rongon**, “*Implementation of Energy Efficiency Model for Cloud based Data Center in Bangladesh*” in IUBAT Innovative Idea Manifest (IIIM 2022) ☐

PROFESSIONAL EXPERIENCE

Tixio Technologies AS <i>Senior Software Engineer (Head of QA and Automation)</i>	<i>Jan 2026 – Present</i>
AskTuring AI Inc. <i>Software Engineer III, AI Model Testing</i>	<i>July 2025 – Dec 2025</i>
Agency Handy <i>Software Engineer II, QA</i>	<i>Aug 2023 – July 2025</i>
Onethread Technologies Limited <i>Software Engineer, QA</i>	<i>Feb 2022 – July 2023</i>

RESEARCH EXPERIENCE

Miyan Research Institute, IUBAT <i>RA – Research Assistant</i>	<i>Jan 2024 – Oct 2024</i>
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TEACHING EXPERIENCE

IUBAT- International University of Business Agriculture and Technology <i>Department of Computer Science and Engineering</i> <i>TA – Teaching Assistant, CSC 231 - Fundamentals of Electronics</i>	<i>Feb 2021 – July 2022</i>
Solution Center of Science and Mathematics <i>Founder and Senior Instructor of ICT</i>	<i>Feb 2019 – Feb 2022</i>

TECHNICAL SKILLS

• Programming & Scripting Languages: Python, C++, MATLAB, SQL • Machine Learning & Data Mining: scikit-learn, pandas, NumPy, TensorFlow. • Research & Simulation Tools: MATLAB, Green Cloud Simulator, CloudSim, NS2 • Data Visualization & Analytics: Power BI, Excel, Google Data Studio, Python Visualization Libraries • UX & Accessibility Evaluation: Heuristic Evaluation, Semiotic Analysis, UX Audits, Figma. • Automation & Quality Assurance: Selenium, Postman, REST Assured, Swagger. • Cloud & Database Technologies: MySQL, MongoDB, AWS (basic) • Version Control & DevOps: Git, GitHub, Jenkins (CI/CD) • Academic Writing & Reference Tools: LaTeX, Overleaf, Microsoft Office	
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SCHOLARSHIPS AND AWARDS

• 1st Runner Up at IEEE SympSIST 2025: For Presenting Poster	<i>Dec 2025</i>
• Research Publication Grant: Awarded by Miyan Research Institute, IUBAT	<i>Dec 2024</i>
• SUSCOM 2022 Conference APC Grant: Awarded by Miyan Research Institute IUBAT	<i>March 2022</i>
• Professor Mahmuda Khanam Scholarship: For Academic performance at IUBAT	<i>Fall 2020</i>
• RL (Bob) Southern Scholarship: For Academic performance at IUBAT	<i>Summer 2021</i>

REFERENCES

Dr. Muhammad Hasibur Rashid Chayon

Associate Professor of Computer Science, International University of Business Agriculture & Technology

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Letter of Recommendation ☑

Mr. Krishna Das

Assistant Professor of Computer Science, International University of Business Agriculture & Technology

Email: krishna.das@iubat.edu

Letter of Recommendation ☑

Lt Col Muhammad Nazrul Islam, PhD

Associate Professor of Computer Science and Engineering, Military Institute of Science and Technology

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